

Topic: String Manipulation

Q. What is a String in Data Structures? Explain the common string manipulation operations with examples.

 Definition to Remember

In C, a **String** is a one-dimensional array of characters terminated by a special **null character** (`'\0'`). String manipulation refers to the set of operations used to modify, analyze, or compare strings. The `<string.h>` library in C provides built-in functions for all major string operations.

1. String Representation in Memory

```
char str[] = "HELLO";
Memory:  H | E | L | L | O | \0
Index:   0  1  2  3  4  5
```

- The **null character** `'\0'` marks the end of the string.
- String size in memory = length + 1 (for `'\0'`).

2. Common String Manipulation Operations

Function	Operation	Description
<code>strlen(s)</code>	Length	Returns number of characters (excluding <code>'\0'</code>)
<code>strcpy(dest, src)</code>	Copy	Copies <code>src</code> string into <code>dest</code>
<code>strcat(dest, src)</code>	Concatenation	Appends <code>src</code> to the end of <code>dest</code>
<code>strcmp(s1, s2)</code>	Comparison	Returns 0 if equal; negative if <code>s1 < s2</code> ; positive if <code>s1 > s2</code>
<code>strrev(s)</code>	Reverse	Reverses the string in place
<code>strupr(s)</code>	Uppercase	Converts all characters to uppercase

3. Code Example

```

#include <stdio.h>
#include <string.h>

int main() {
    char str1[20] = "Hello";
    char str2[20] = "World";
    char dest[40];

    // 1. Length
    printf("Length: %lu\n", strlen(str1));           // Output: 5

    // 2. Copy
    strcpy(dest, str1);
    printf("Copy: %s\n", dest);                     // Output: Hello

    // 3. Concatenation
    strcat(dest, " ");
    strcat(dest, str2);
    printf("Concat: %s\n", dest);                   // Output: Hello World

    // 4. Comparison
    if (strcmp(str1, str2) == 0)
        printf("Equal\n");
    else
        printf("Not Equal\n");                     // Output: Not Equal

    return 0;
}

```

★ Must-Write Points (for 10 marks)

1. A string in C is a character array terminated by **null character** `'\0'`.
2. String manipulation uses the `<string.h>` library in C.
3. `strlen(s)` → counts characters excluding `'\0'`.
4. `strcpy(dest, src)` → copies one string to another.
5. `strcat(dest, src)` → appends src to the end of dest.
6. `strcmp(s1, s2)` → returns 0 if equal; lexicographic comparison.
7. Strings are used in text processing, data validation, and pattern matching (KMP, Rabin-Karp).

⚡ Quick Recall

String → char array + `'\0'` → `strlen` (length) → `strcpy` (copy) → `strcat` (join) → `strcmp` (compare, 0=equal) → `strrev` (reverse) → `<string.h>`